CMSC 35200

Deep Learning Systems

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2026 introduction · DRAFT · 27 September 2026

The 2026 course

Understand the models and the systems that make them useful.

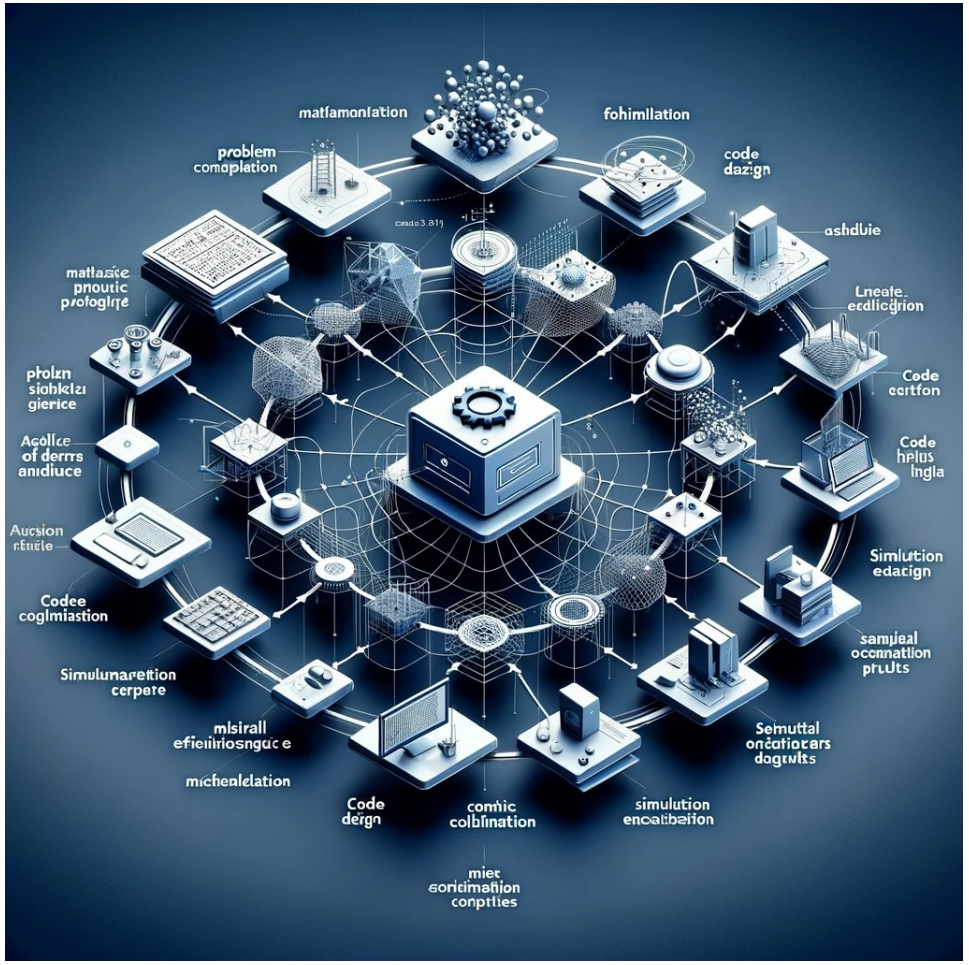
Build with coding agents, then verify the result independently.

Measure accuracy, latency, memory, concurrency and cost.

Use scientific problems to test the limits of scalable cognition.

AI Assistant for “Computational” Science

- Open Problem
- Mathematical Formulation
- Numerical or Symbolic Formulation
- Efficient Algorithm
- Parallel HPC implementation
- Simulation Campaign at Exascale HPC
- Analysis of Output
- Insights and Closing the Loop



GPT4 Notion of what AI assistant Looks like

Research ambitions and testable claims

Scientific discovery motivates the systems we will build.

A compelling demonstration needs a falsifiable claim and a baseline.

The next slides describe a research agenda.

Our projects will test a bounded part of that agenda.

What it might take to get to a **“scientific superintelligence”** that can run for days or weeks without intervention, master the literature, design instruments, direct large robotic facilities, and originate new theories.

The scientific discovery cycle

Implementation synthesis based on DOE’s closed-loop vision

01 **Frame the question**

Set the scientific objective, constraints and success criteria.

02 **Assemble evidence**

Retrieve relevant observations, theory and prior experiments.

03 **Propose and predict**

Generate hypotheses and estimate which tests will teach the most.

04 **Run the test**

Execute approved simulations or physical experiments.

05 **Validate the result**

Compare prediction with measurement and quantify uncertainty.

06 **Update and repeat**

Preserve the evidence and use it to choose the next question.

Synthesis informed by DOE National Challenges and Models teams

A portfolio of scientific models

Different scientific tasks need different model capabilities.

Reasoning models

Interpret questions, assemble evidence and plan scientific work.

Domain foundation models

Represent scientific structures and relationships in specialized data.

Surrogates and predictors

Approximate costly calculations and estimate promising candidates.

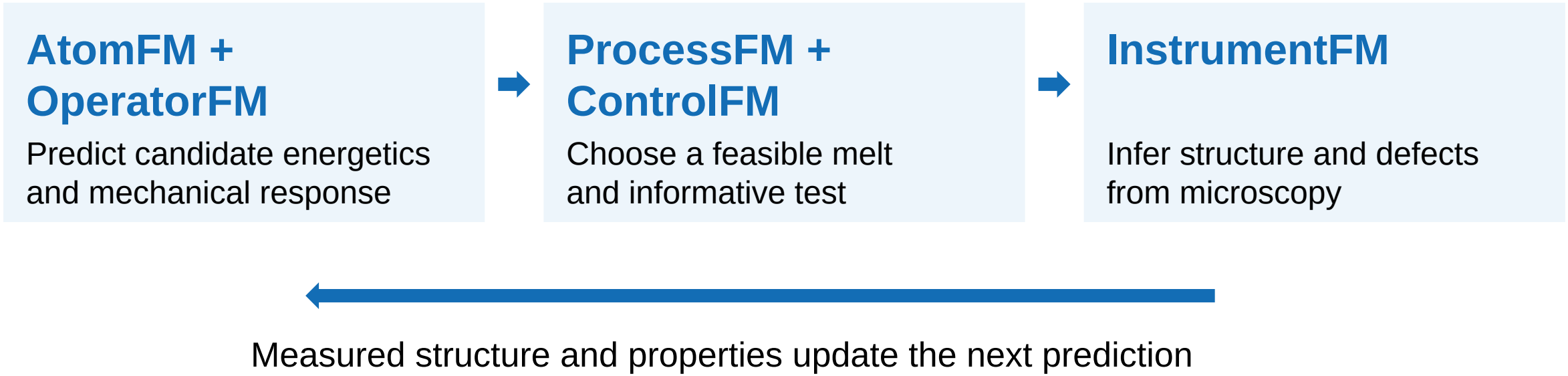
Evaluation and model registry

Track capabilities, limitations, versions and evidence for reuse.

Open-weight scientific models add a path for community adaptation and scrutiny.

A discovery requires models to compose

Question: which oxygen-containing alloy should we make next, and why?



CodeFM executes simulations. EvidenceFM records the chain of justification.

Course roadmap

Weeks 1–2 Coding loops, tensor notation and transformer foundations

Weeks 3–4 Attention variants, multimodal models and scaling

Weeks 5–6 Inference systems, concurrency and resource planning

Weeks 7–8 Agent evaluation, decision models and scientific computing

Weeks 9–10 Scalable cognition, replication and project defenses

Key Transformer Paper

arXiv:1706.03762v5 [cs.CL] 6 Dec 2017

Attention Is All You Need

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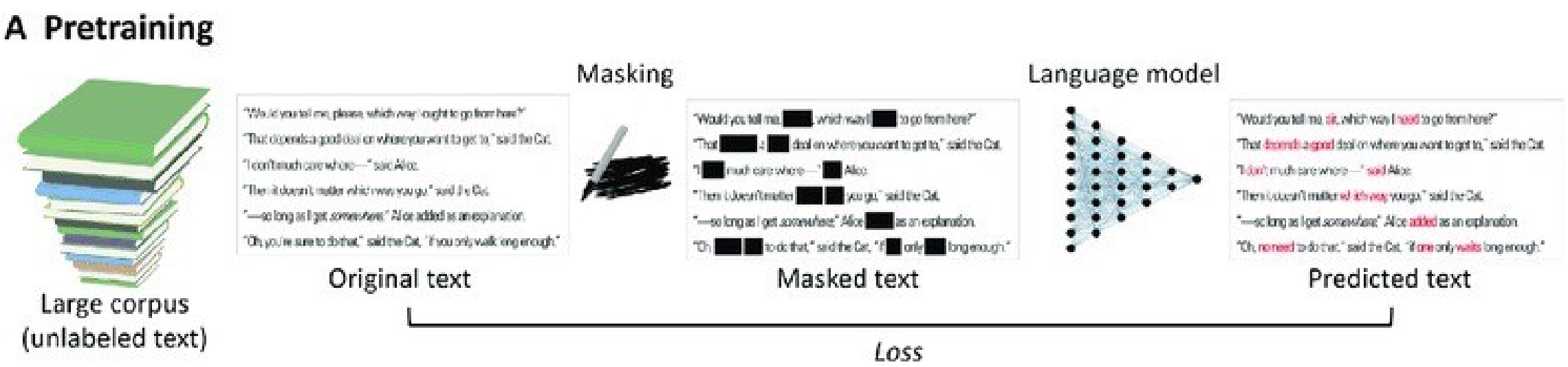
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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Self-Supervised Pretraining and Fine-Tuning



Tensor notation as an executable specification

$$Q_{bthk} = X_{bti}\ W_{ihk}$$

$$S_{bhts} = Q_{bthk}\ K_{bshk}\ /\ \sqrt{D}$$

$$A_{bhts} = \mathrm{softmax}_s(S_{bhts} + M_{ts})$$

$$O_{bthv} = A_{bhts}\ V_{bshv}$$

Sum contracted indices i, k and s. D is the key width.
Free indices preserve batch, position and head structure.

Attention variants and memory

Multi-head attention provides the reference implementation.

GQA shares key/value heads across groups of query heads.

MLA compresses the representation retained in the KV cache.

Linear and gated recurrent attention change the state and computation.

LLM and multimodal scaling

Training scale includes parameters, tokens and data quality.

Inference scale includes context, sampling and reasoning budgets.

Images, video and scientific arrays add representation and alignment costs.

The experiment must report capability together with memory and time.

Jev as a decision-model case study

Jev exposes typed decisions rather than free-form answer generation.

Compare decision quality and calibration with a language-model baseline.

Measure latency, abstention and the cost of a wrong decision.

Valid output structure still requires a correct decision.

Agents turn capabilities into scientific work

The research task is a persistent stateful campaign, not a sequence of isolated prompts.

Reason and plan

Track hypotheses, competing explanations and uncertainty; choose the next simulation or experiment.

Evidence and simulator grounding

Execute and recover

Call domain models, schedule ensembles, use instruments and recover from routine failures.

Typed tools and durable task state

Verify and govern

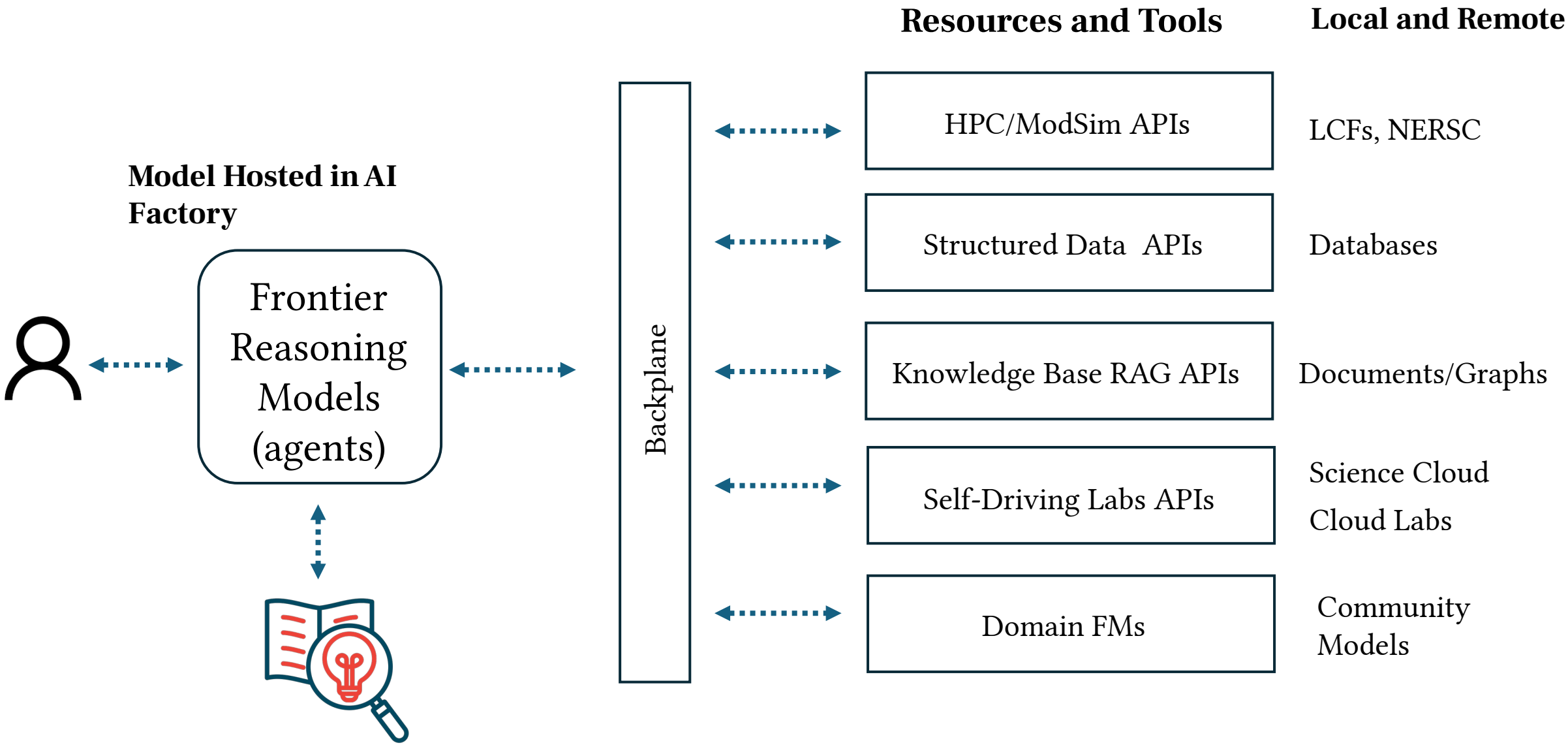
Check physical constraints, preserve provenance and route consequential actions through defined authority.

Runtime monitoring and escalation

Multi-day autonomy requires state, budgets, boundaries and recovery.

Source: ModCon notes + projects 297659 and 301923 • Synthesis

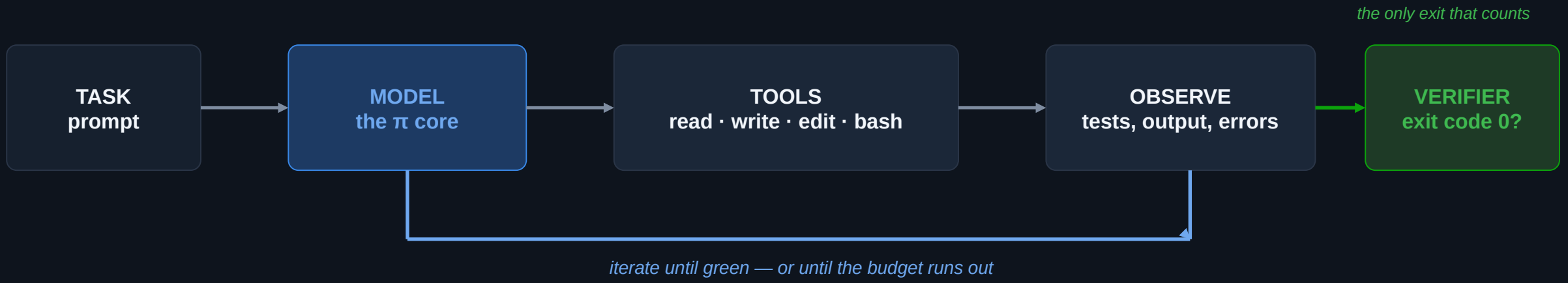
Scientific AI Platform Example



THE BIG IDEA

Models aren't chatbots here — they're loop cores

An agent loop: the model reads a task, acts through tools, observes reality, and iterates until an independent judge says done.



file-path discipline

artifacts must land where the grader looks

recover from red

read a pytest traceback, make the smallest fix

drive multi-step tools

no wandering, no giving up early

know when it's done

stop on green — don't 'improve' passing code

A working coding loop on every laptop

A reproducible Python environment and a clean Git checkout

A coding agent that can read, edit and run the project tests

One approved model endpoint and an offline test path

A failing seed, a verified repair and a recorded run

THE BENCHMARK

pi-30 in one slide

1 problem = seed code (pre-verified broken) + task prompt + loop budget + **verifier (exit code = verdict)**

30 problems

P1–P30, authored as a curriculum: write-from-spec → fix-failing-tests → numerical → optimize → tournaments & critics.

1 harness

run_model_30.sh — problem statements embedded, every pi call timeout-wrapped, resume-safe, fleet-parallelizable.

1 lane = model × backend

The same weights on two serving stacks are two lanes. 26 unique lanes have completed the sweep.

1 verdict per problem

Binary PASS/FAIL from an independent grader the model may not edit. Score = passes out of 30. No partial credit, no vibes.

STANDING RULE 1 / 6

The verifier is the only source of truth

- Model prose is never trusted for scoring. Not “done!”, not “all tests pass”, not a beautiful summary.
- Each problem ships an independent grader: verify.py, check.py, pytest, a benchmark scorer, or a convergence check.
- The grader's process exit code is the verdict: 0 = PASS, anything else = FAIL.
- The model is forbidden from editing the grader — every prompt says so, and staging restores pristine copies each run.

run_model_30.sh + p1/verify.py

```
# harness verdict line — the whole religion:
(cd "$ROOT/p1" && "$PY" verify.py) \
  >/dev/null 2>&1 \
  && record "P1: PASS" \
  || record "P1: FAIL"

# verify.py ends with hard asserts:
assert rc == 0
assert abs(d["stdev"] - 2.138089935) < 1e-6
print("OK P1")   # exit 0 = the verdict
```

Why so strict? agent models are trained to sound finished. An eval that reads prose measures persuasion; an eval that runs code measures capability. (You'll see models claim victory over code that doesn't import.)

ANATOMY

Inside one problem directory

problems/p24 (token bucket)

```
problems/p24/
├─ limiter.py      ← SEED (broken on
│                     purpose; must fail)
└─ verify.py       ← GRADER (never
                     edited by the model)

# graders across the suite:
verify.py          behavior asserts
check.py           oracle diff
test_thing.py      pytest suites
bench.py + check.py  speed + correctness
score.py           tournament scorer
converge_check.py  numerical convergence
PLAN.md / CRITIQUE.md  fan-out & critic aids
```

The staging contract (per lane)

- rm -rf runs30/<tag> — a lane never sees a previous lane's artifacts
- Fresh copies of every verifier + seed staged into runs30/<tag>/pN/
- pi runs with cwd = the problem dir; the grader runs from the same dir moments later
- Verifiers execute under a pinned Python (numpy + pytest available) — not whatever the model's bash found on PATH
- “DO NOT edit check.py/verify.py” appears in every prompt — and staging would clobber the edit anyway

Backend access is part of the system

Record provider, endpoint, model ID, version and serving configuration.

Check authentication, tool calls, timeouts and rate limits before grading.

Set token and spending budgets. Keep credentials outside the repository.

Separate infrastructure failures from task failures.

The benchmark contract

Freeze tasks, held-out tests, prompts and resource budgets.

Compare systems on the same tasks with independent repeats.

Report success rate, failure classes, latency and cost per valid result.

Keep raw artifacts and distinguish first attempts from retries.

Concurrency changes the experiment

Independent trials can run in parallel. Shared state introduces contention.

Queues, retries and cancellation affect measured throughput.

More agents can duplicate work and amplify correlated errors.

Compare serial and concurrent runs at a fixed total budget.

SCALE OUT

Fleet mode: 13 lanes in parallel, politely

harness/fleet_launch.sh (condensed)

```
MODELS=(gemma4-e2b gemma4-e4b gemma4-12b
  gemma4-31b ornith-9b devstral-small-2
  qwen36-27b uic-ornith-9b … )  # 13 total

for m in "${MODELS[@]}"; do
  PI30_RESUME=1 PI_TIMEOUT=360 \
  setsid bash run_model_30.sh \
    "$PROVIDER" "$m" "fleet-$m-$STAMP" \
    > "$log" 2>&1 &
  setsid bash -c "… heartbeat …" &
done; wait

# setsid: each lane gets its OWN process group —
# SIGTERM to the launcher cascade-kills NOTHING
```

Fleet furniture

- One aggregator provider (chiatta00-agg) fronts all 13 models — the launcher just varies --model
- Heartbeat sidecar per lane: greps the latest [Pn] marker + pass/fail counts into <model>.cur every 5 s
- PI30_RESUME=1 by default — a re-launch continues every partial lane instead of burning finished work

Architecture and resource planning

Estimate weight storage, KV state, activations and runtime overhead.

Separate prefill, decoding and tool execution in the time model.

Plan training, serving and agent experiments as different workloads.

Use measured utilization and a sensitivity analysis.

The hypothetical 10T MoE workload

Base case: 10T total parameters, 500B active per token, 20T training tokens.

Quantity	Planning assumption	Implication
Training duration and precision	100 days, FP8 mixed precision	BF16/FP32 state retained
Training model state	16 bytes/parameter + 25% reserve	200 TB sharded memory floor
Inference weights	FP4 + 10% quantization metadata	5.5 TB resident weights
Inference context and batch	8,192 input tokens, 64 sequences	137 GB KV cache at assumed layout
Output and service demand	1,024 output tokens, 10K output tok/s	Separate decode and prefill energy

These are analyst assumptions. 20T training tokens is a scenario, not an assertion of optimal model quality.

A small planning exercise

Illustration: 20 tasks × 3 repeats = 60 trials

At 8,000 input + 2,000 output tokens per trial: 600,000 tokens

Price input and output separately using the approved endpoint rate.

Measure one trial before selecting concurrency or committing the budget.

A theory of scalable cognition

Working question: when does additional computation improve verified work?

Vary model capability, memory, tools, search depth and parallel effort.

Track useful progress per unit of time, tokens and energy when measurable.

Test limits from coordination, correlated errors and weak verification.